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Name2Vec: Name Matching using Character-based with Deep Learning

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Abstract

Name matching plays a crucial role in big data and various integration applications, being indispensable when consolidating information from diverse sources. This encompasses tasks such as deduplication, data linkage systems, search engines, text and web mining, information extraction, and more. Discrepancies and anomalies in names, including syntax variations like abbreviations, typographical errors, occasional whitespace omissions, word insertions, deletions, and even multiple spellings for the same name, can lead to missed matches. In previous methodologies, a predefined penalty scheme was often employed for each differing character or multi-character token between two strings. This research introduces Name2Vec, an algorithm that addresses name matching using a neural network model to capture name semantics. This approach advances by suggesting a suitable feature set through the fusion of Name2Vec and character-based name representations. The empirical findings of this research confirm that this performance enhancement improves matching efficiency while simultaneously reducing misclassifications compared to state-of-the-art methods.

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Keywords: Name Matching String Similarity, Character-Base, Name2Vec, Deep Learning.

1. Introduction

The Internet currently grants access to vast repositories of nominal data and knowledge, particularly in the realm of Big Data [1]. Names hold paramount importance across diverse societies, including those at the forefront of technological advancement [2]. Name matching entails the pivotal task of definitively ascertaining if two distinct representations correspond to an identical real-world entity. This challenge persists prominently across numerous organizations and integrated Big Data applications, where data is scattered across heterogeneous and disparate sources, necessitating the identification and management of duplicate entries [3, 4, 5].

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The term *name matching* also loosely encompasses the broader challenge of discerning whether two heterogeneous representations of distinct individual names should be correlated. The identification and alignment of personal names constitute a pivotal component within an expanding spectrum of applications, ranging from deduplication and knowledge linkage systems to search engines, data extraction, and text and web mining, all within the context of Big Data [5, 6]. Names, particularly surnames, occupy a paramount role as names find extensive utility in various processes, including the differentiation of individuals, genealogical analysis, the compilation of birth and death records, parish registries, textual articles, and multimedia systems [2, 6]. Effectively harnessing these data resources necessitates multifaceted linkages, considering factors such as familial name associations, name similarity, shared spatiotemporal locations, and legal affiliations [7].

The challenge associated with name matching primarily arises due to the absence of standardized protocols for systematically storing such entities and their distinctive identifiers [8]. Variations and errors in names manifest when records pertaining to the same individual exhibit syntactical differences, including abbreviated forms, typographical errors, sporadic whitespace omissions, word insertions, deletions, and the existence of multiple spellings for identical names, all of which can lead to inconclusive matches [9]. Conversely, variations in names can pose a significant hindrance in the context of identification and search processes, such as web searches or security-related tasks. Name matching presupposes that the recorded name is represented in a single alphabet corresponding to its phonetic identity [2]. Name-matching methodologies serve to quantify the level of correspondence between individual names and furnish a score indicative of the match's strength. A higher match score implies a greater likelihood that the records pertain to the same individual. However, numerous existing name-matching techniques tend to be generic and language or culturally agnostic. Consequently, while a specific method may yield optimal results in one country or culture, its performance may not be consistently superior in another [10].

A straightforward illustration of the challenges encountered in matching an identical name across two information sources with differing formats is exemplified by the fact that in one case, an individual's name may be stored in the database using only two columns, one for the family name and another for the full name. In contrast, the second database may include columns for the last name, middle name, first name, first name abbreviations, titles, and honorifics. These variations and errors in name structure pose challenges for exact string matching, necessitating the application of approximate matching techniques. In comparison to generic textual data, personal names exhibit distinct characteristics that demand specialized consideration [6]. The prevalent approaches for addressing name matching dilemmas encompass the utilization of phonetic encoding and pattern matching. Furthermore, additional techniques have been devised for each of these approaches, some even combining aspects of both to enhance the quality of matching. These techniques rely on similarity measures, as opposed to exact identity matching, incorporating methods such as edit distance, n-gram matching, and Soundex [10]. Despite substantial research efforts in this domain, phonetic matching methods like Soundex, Metaphone, and Double Metaphone have primarily been developed with a focus on phonetic characteristics specific to English names [6, 10]. Pattern matching techniques, such as Edit Distance, are language-independent but often prove to be too general, failing to capture the full spectrum of possible variations specific to distinct cultures or countries [10].

Previous methodologies imposed predetermined penalties for each character or multi-character token that differed between two strings [11]. Much of the research in personal name matching has centered on Indo-European languages, notably English, where characters or character combinations correspond to phonemes, and the character set is relatively limited. These characteristics facilitate direct comparisons of character string data, yielding reasonably accurate measures of similarity in terms of phonetics, visual appearance, and keystroke sequences [12, 13]. To accommodate the variability that can arise due to errors in the data entry process and the diverse representations of the same concept or entity, such as abbreviations, transliterations across languages, nicknames, and rearrangements of given and family names, multiple heuristic techniques have been developed to evaluate the similarity of character strings. Conventional methods like cosine similarity and Levenshtein similarities impose fixed penalties for each individual character or multi-character token that distinguishes one string from another [6, 13, 15]. In contrast, learnable similarity metrics can be trained using labeled samples of co-referent and non-co-referent names. Similarity metrics can be acquired through machine learning classifiers trained on features derived from pairwise string comparisons, encompassing static similarity scores [3, 16], and probabilistic finite-state transducers, which estimate the probability of specific variations within the context of adjacent characters [17, 18, 19]. Training classifiers with multiple static similarity metrics can unveil nonlinear relationships between similarity scores and matching probabilities, offering a

data-driven approach to weighing various definitions of string similarity. Finite-state transducers, on the other hand, can learn penalties associated with specific character deletions, insertions, and substitutions within their contextual surroundings. However, these transducers are notably more intricate and necessitate extensive training and validation datasets to achieve robust performance [13].

This research delves into the intricacies of personal names, elucidating the underlying sources of variability and potential errors. Subsequently, it presents a comprehensive survey of conventional methodologies, as well as recent advancements in name matching techniques. This study embarks on innovative strategies aimed at augmenting the precision of name matching by fusing character-based similarity methodologies with deep learning techniques. This paper introduces Name2Vec, a pioneering solution for name matching, which harnesses the power of deep learning models to discern the semantics of names. Additionally, this research proposes a suitable feature set by amalgamating Name2Vec with character-based representations. Empirical validation of this approach reveals significant performance enhancements, making name matching more effective and reducing misclassification rates when compared to prevailing state-of-the-art methods

2. Related works

Name matching, as a research domain, has witnessed a surge in interest due to its critical role in numerous real-world applications, including data integration, deduplication, and information retrieval. Researchers have recognized the pivotal importance of accurately aligning names, which often serve as the key identifiers in heterogeneous datasets. While prior research has largely emphasized pairwise schema or ontology comparisons, it is important to note that the challenges associated with name matching extend far beyond these comparisons. Variability in name representations, such as abbreviations, multiple spellings, and linguistic variations, further complicates the task and necessitates the exploration of innovative methodologies. In addition to supervised, unsupervised, and semi-supervised approaches, recent advancements have also introduced hybrid techniques that combine the strengths of multiple methodologies. These hybrid approaches aim to provide more versatile and adaptive solutions for addressing the complexities of name matching in diverse data environments.

Within the realm of unsupervised approaches, the utilization of clustering algorithms plays a crucial role in the efficient organization of vectors, facilitating the identification of inherent patterns and relationships among data points. These unsupervised methodologies provide a versatile framework for addressing the complexities of name matching across diverse datasets. Character-based similarity metrics delve into the intrinsic properties of individual characters, token-based similarity metrics consider entire tokens or sub-strings as units of comparison, while phonetic similarity metrics transcend character-level analysis to incorporate phonetic attributes, offering a comprehensive array of techniques for tackling the multifaceted challenges posed by name matching.

Character-based similarity metrics have been devised specifically to address typographical errors encountered effectively in name matching. These techniques include a variety of metrics, such as Edit distance, Affine gap distance, Smith-Waterman distance, Q-gram distance, Levenshtein distance [14], Jaro distance [23], and Winkler distance [23]. Additionally, the Fuzzy-Wuzzy package [24], which relies on the Levenshtein Distance algorithm to calculate dissimilarities between distinct strings, has displayed strong performance. This renders it a valuable tool in the domain of name matching, both for individual words and significantly longer texts.

Token-based similarity metrics prove highly effective when dealing with errors involving word rearrangement. This strategy entails the division of two string sequences into word sets, followed by the calculation of their similarity. Prominent techniques in the domain of token-based similarity metrics encompass the application of atomic strings originally proposed by Monge and Elkan [25], the WHIRL method [23], the TF-IDF metric, and the utilization of Q-grams in conjunction with TF-IDF [26].

Phonetic similarity metrics opresent solutions in scenarios where strings may exhibit phonetic resemblance, even when disparities exist at the character or token level. These metrics encompass methodologies such as Soundex [27], the NYSIIS system [28], ONCA [29], as well as Metaphone and Double Metaphone [28]. These phonetic-centric approaches offer valuable resources for addressing situations where pronunciation similarity takes precedence over strict character or token alignment.

Supervised approaches in name matching rely on labeled training datasets or predefined thresholds to guide their decision-making process. Name matching is frequently conceptualized as a classification task [10], where each pair of

names serves as an instance, and a positive classification signifies that two names refer to the same entity. When candidate name pairs are categorized into matches and non-matches, this classification can be viewed as a binary classification problem, well-suited for supervised classification techniques. However, practical applications often grapple with the challenge of determining sample variations within training datasets, which can impose limitations on the effectiveness of these solutions. In the realm of supervised learning for name matching, there exist two prominent approaches: Character Embedding and Text Embedding [21, 22]. Text embedding entails the transformation of words into numerical vectors, capturing grammatical functions, syntax, and word semantics. This facilitates the execution of various mathematical operations on these vectors, enabling deeper analysis of word relationships. Conversely, character-level embedding leverages a one-dimensional convolutional neural network to generate numeric representations of words based on their character-level compositions. Character embedding proves particularly advantageous when dealing with out-of-vocabulary words, a challenge that text embedding may struggle to address effectively. These supervised learning methodologies play a pivotal role in enhancing the accuracy and robustness of name matching systems, especially in scenarios where precise entity recognition is crucial.

Semi-supervised approaches amalgamate both unsupervised and supervised methodologies, incorporating unsupervised techniques without the need for labeled training data, and harnessing supervised methods with access to labeled training data. These semi-supervised strategies rely on a relatively small set of labeled data in conjunction with a substantially larger set of unlabeled data. This research advocates for the adoption of word embeddings for handling names and the integration of property values as part of this approach. Furthermore, semi-supervised techniques offer a pragmatic solution for scenarios where obtaining labeled training data on a large scale is prohibitive. By leveraging the vast reservoir of unlabeled data, these approaches can significantly enhance the robustness and scalability of name-matching systems, providing valuable insights into entity recognition and disambiguation.

3. Methodologies

In this research, a comprehensive approach is employed, consisting of four distinct stages, as visually represented in Figure 1:

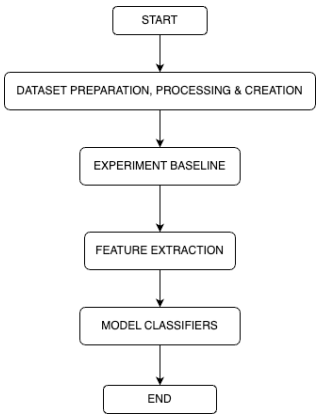

Fig. 1. Experiment Scenario

3.1. Dataset preparation, processing, and creation

This research endeavors focus on the outcomes of an extensive series of comparative trials conducted using a Vietnamese name dataset, with a primary aim to evaluate the performance of various name-matching techniques across diverse categories of personal names. Additionally, this research assesses the computational efficiency of these techniques. To facilitate these investigations, this research has implemented all name-matching methodologies in Python, seamlessly integrating them into the Febrl data binding system (Freely Extensible Bio-medical Profile Linking) [5],

a comprehensive system designed for managing and processing large-scale data, including Big Data. The Febrl system encompasses three critical stages: Data Processing, Model Building, and Testing. The initial phase of this study entailed the collection of Vietnamese names from two online sources, obtained through web crawling techniques.

The extensive dataset used in this study consisted of 24,210 Vietnamese personal names, specifically obtained from high school students' records through web crawling techniques and sourced from Facebook data. This substantial dataset formed the cornerstone of this research, providing a robust foundation for investigations of this research into name matching. Following the acquisition of the dataset, this research conducted a thorough and meticulous data preprocessing phase. This essential step is critical when dealing with large datasets and encompasses various procedures. These procedures include the removal of extraneous whitespace, the elimination of honorifics and titles, as well as the careful handling of duplication and missing values within the dataset. This data preprocessing phase ensured that dataset was clean and ready for subsequent analysis.

To facilitate binary classification tasks, this research thoughtfully structured the dataset into two distinct categories: positive and negative examples (instances). Positive examples represented pairs of names that were determined to be matching or similar, while negative examples encompassed dissimilar pairs of names. This categorization allowed us to formulate meaningful classification problems and evaluate the effectiveness of this name-matching techniques. The diligent data set generation stage aimed to create a well-balanced dataset, striking a harmonious balance between positive and negative examples. This balanced dataset was crucial for the reliability and robustness of experiments, as it provided an accurate representation of real-world scenarios in name-matching applications. This research embodies a comprehensive and methodical approach, underpinned by rigorous methodologies. This commitment extends to advancing the field of name matching, particularly within the context of Vietnamese personal names, while also contributing to the broader domain of data-intensive applications, including the realm of Big Data analytics which is shown in figure 2.

	name	fb_name	Label
0	anh bình	Trương Quốc Bình	0
1	anh bình	Bi Bo	0
2	anh chinh	Trường Chinh	1
3	anh chinh	Cherry Nguyen(Ngọc Quý)	0
4	anh chinh	anh chinh	1
...	...	...	...
19411	yên phúc hải	Chơn Viên	0
19412	yên phúc hải	yên phúc hải	1
19413	yên thị muối	Nga Kim	0
19414	yên thị muối	Loan Le(Lynh)	0
19415	yên thị muối	yên thị muối	1

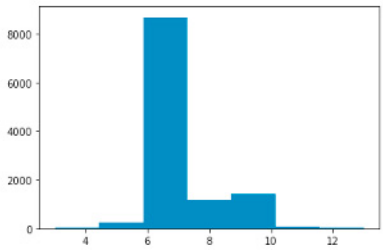

Fig. 2. Data preparation result and Token count statistics for the set of names

Vietnamese datasets often exhibit a high degree of noise and contain numerous errors. These inherent challenges can significantly impact the performance of classification models. Therefore, a crucial initial step in this research is diligent data preprocessing to enhance data quality. Data preprocessing involves a series of essential steps designed to clean and prepare the dataset for subsequent classification tasks. These steps encompass the removal of special characters, the normalization of hashtag text to lowercase, and the exclusion of excessively short sentences. By implementing these measures, this research aims to create a cleaner and more reliable dataset, which is foundational for achieving accurate results in classification. In summary, this research emphasizes the critical role of data preprocessing in mitigating the challenges posed by noisy Vietnamese data. These preparatory steps are fundamental to ensuring the effectiveness and reliability of this classification model when working with such datasets.

3.2. Experiment baseline

Within the context of this scientific research investigation, a methodological framework was founded as the baseline of the experiment using Machine Learning classification methods. This foundational approach is driven by the overarching objective of establishing a performance baseline that can be juxtaposed with our novel and innovative methodol-

ogy. This research has harnessed the computational power of three distinct algorithms, namely Linear Support Vector Machine (SVM), Logistic Regression (LR), and Random Forest (RF), to facilitate the comparative analysis.

In the assessment of name matching, a comprehensive evaluation is conducted, wherein each method's performance is scrutinized in relation to individual optimal threshold values. Pairings exhibiting scores below the stipulated threshold are definitively categorized as non-matching pairs, while those that surpass the threshold are unequivocally classified as matching pairs. Subsequent to this critical classification process, this paper undertakes a meticulous examination of the outcomes, juxtaposing them with the ground truth labels assigned to each pair. This meticulous procedure facilitates the quantification of accurately classified pairings as well as those erroneously categorized.

To determine the overall effectiveness of each employed algorithm, this research relies on the F1 measure metric as a pivotal evaluation tool. This meticulously applied metric allows for the discernment of the predictive capabilities inherent in the tested algorithms. Furthermore, as this paper delves deeper into the intricacies of this scientific exploration, this paper can extend this analysis to shed light on the broader implications and significance of the findings, thereby enriching the understanding of the subject matter.

3.3. Feature Extraction

The data construction process involves generating characteristic indexes for two series. These feature indexes are computed using seven distinct extraction algorithms, namely Jaro Distance, Jaro Winkler, FuzzyWuzzy, LCS (Longest Common Subsequence), LCSubStr (Longest Common Substring), Sequence Matching, and Jaccard similarity. The figure 3 below illustrates the results of this data construction, providing a visual representation of the computed feature indexes. The presented figure offers a comprehensive view of the characteristic indexes obtained through the application of these seven extraction algorithms. Each algorithm contributes unique insights into the data, allowing for a detailed analysis of the constructed feature indexes. This data construction phase plays a pivotal role in this research, as it lays the foundation for subsequent analyses and evaluations. The diverse set of extraction algorithms employed in this process ensures a robust exploration of the dataset, facilitating a deeper understanding of the characteristic indexes and their relevance to this study.

	label	name	jaro_distance	jaro_Winkler	fuzzywuzzy	LCS	LCSubStr	sequencematching	jaccard
0	0	anh binh & trương quốc binh	0.46	0.46	0.35	0.38	0.31	0.50	0.25
1	0	anh binh & bi bo	0.55	0.55	0.31	0.25	0.25	0.31	0.00
2	1	anh chinh & trương chinh	0.79	0.79	0.45	0.58	0.50	0.67	0.40
3	0	anh chinh & cherry nguyen ngọc quý	0.59	0.59	0.25	0.14	0.09	0.19	0.06
4	1	anh chinh & anh chinh	1.00	1.00	0.67	1.00	1.00	1.00	1.00
5	1	anh tạo & tao nguyen	0.74	0.74	0.33	0.20	0.10	0.24	0.00
6	0	anh tạo & sy nguyen	0.50	0.50	0.24	0.11	0.11	0.12	0.00
7	1	anh tạo & anh tạo	1.00	1.00	0.36	1.00	1.00	1.00	1.00
8	1	anh trí & tri nguyen	0.65	0.65	0.50	0.20	0.20	0.24	0.10
9	0	anh trí & huy vo khac	0.49	0.49	0.24	0.18	0.09	0.22	0.00
10	1	anh tuấn & tuan vu	0.69	0.69	0.50	0.50	0.25	0.40	0.29
11	0	anh tuấn & khoa nguyen	0.76	0.76	0.40	0.36	0.09	0.32	0.00
12	1	anh tuấn & anh tuấn	1.00	1.00	0.46	1.00	1.00	1.00	1.00
13	0	âu dương mỹ yến & mei yan ou yang	0.60	0.60	0.27	0.40	0.13	0.27	0.08

Fig. 3. Data processing of similar feature indexes of name set

3.4. Experimental model

This research primarily focuses on the task of name matching using character-based methods enhanced by deep learning techniques, specifically leveraging name embedding and word embedding methodologies. The overarching goal of this investigation is to maximize the incorporation of sequence-related information into various embeddings,

Fig. 4. Experimental model Attrbuted character token and name embedding

which serve as crucial inputs to this model. This strategic approach is paramount in ensuring that model possesses the highest probability of correctly attributing labels to sequences, thereby optimizing the model's output.

Figure 4 illustrates the architecture of proposed model , highlighting distinctive approach of this research. Instead of employing GloVe embeddings, this research has opted for pretrained TensorFlow Hub token embeddings, a decision that significantly impacts the model's performance. Moreover, this research has introduced a Dense layer atop token-character hybrid embeddings, deviating from the conventional bi-LSTM layer commonly used in similar applications. This design choice has a profound effect on the training process, as it necessitates the continuous updating of both token and character embeddings throughout training. Notably, the pretrained TensorFlow Hub embeddings remain static, offering stability to certain aspects of proposed model.

In terms of optimization, this research adopts a dual optimizer strategy, employing both the SGD and Adam optimizers. This hybrid approach augments the efficiency and effectiveness of this training regimen, contributing to superior performance in sequence labeling tasks.

Additionally,the work integrated name2vec, a novel technique for encoding semantic information within names, into overall model architecture. This name embedding methodology enhances the model's ability to discern subtle semantic nuances, facilitating improved name matching accuracy.By delving into the intersection of name matching, character-based techniques, deep learning, and embedding methods, this research strives to advance the state-of-the-art in the field. This research prioritize empirical validation, underpinning choices with rigorous experimentation and data-driven insights.

In summary, this research underscores the importance of name embedding, word embedding, and name2vec techniques in the context of name matching. The unique model architecture, optimizer strategy, and embedding methodologies collectively contribute to the advancement of name matching using character-based methods enhanced with deep learning techniques.

4. Experimental Results

Upon conducting experiments with each method, utilizing both baseline and machine learning classification approaches, the results can be compared by examining the data below: Upon a comprehensive analysis of the acquired results, the figure 5,6,7,8,9 and 10 evident that the implementation of Attributed Character Token and Name Embedding techniques exhibits remarkable performance, achieving an impressive accuracy rate of 97.6%. In direct comparison, this surpasses alternative methodologies, including Hybrid Character Embedding, CNN Character Embedding,

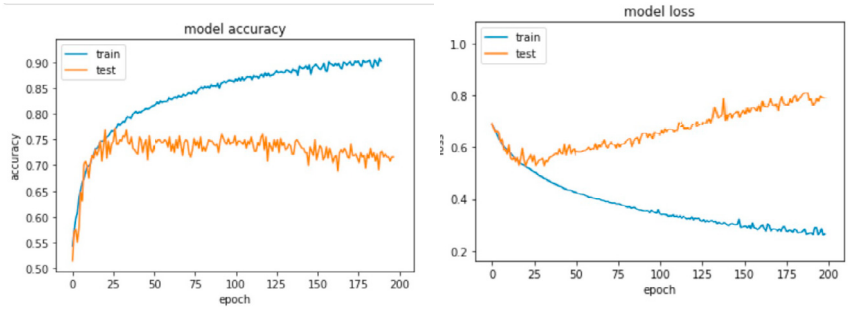

Fig. 5. Attribute and Token Name Embedding CNN Model Results

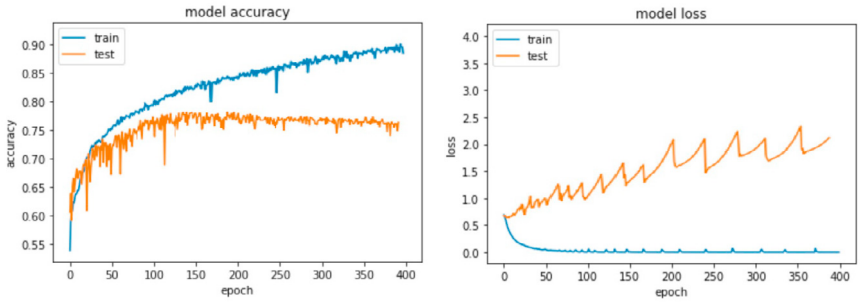

Fig. 6. Token Name Embedding LSTM Model Results

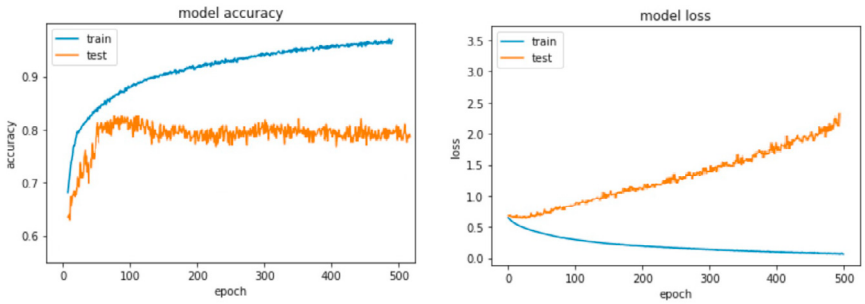

Fig. 7. Attribute and Character Name Embedding CNN Model Results

LSTM Name Embedding, CNN Name Embedding, and Attribute Character LSTM Name Embedding approaches. This observation aligns with the prevailing notion that leveraging individual character-level features can yield more favorable outcomes than the amalgamation of name-based data and the incorporation of attributes obtained from string comparisons.

The remarkable success achieved by the Attributed Character Token and Name Embedding method underscores the significant potential of name embedding techniques in elevating the precision of name matching tasks. Moreover, it highlights the pivotal role played by character-level information in effectively addressing this intricate task. These groundbreaking findings contribute significantly to the expanding landscape of research on Name Matching Using Character-Based with Deep Learning, shedding light on the promising prospects offered by name2vec methodologies and word embedding strategies for achieving robust results within the domain of name matching

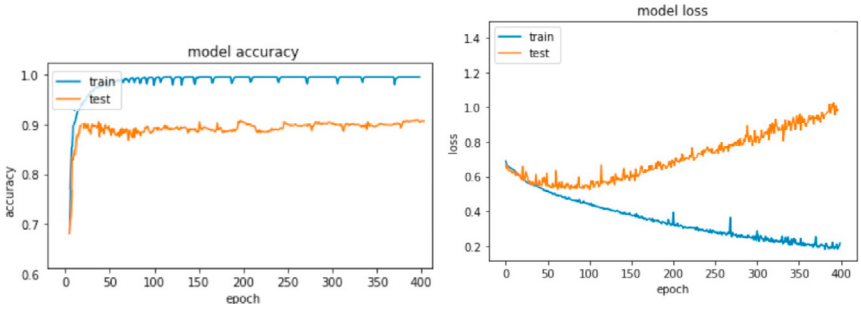

Fig. 8. Character and token Name Embedding CNN Model Results

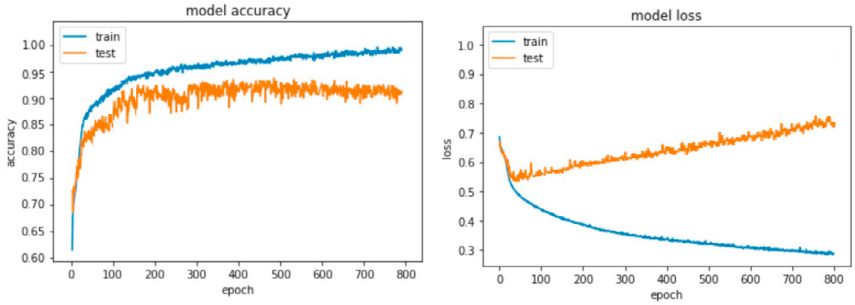

Fig. 9. Attribute and Character, Token Name Embedding CNN Model Results

	accuracy	precision	recall	f1
baseline	72.183238	0.718647	0.721832	0.698925
attri_token_embed_conv1d	78.448961	0.781494	0.784490	0.782299
token_name_embed	81.425261	0.814881	0.814253	0.811455
attri_char_embed_conv1d	91.877797	0.854501	0.858778	0.851686
hybrid_char_token_embed	93.623064	0.917075	0.916231	0.913561
attri_char_token_embed	97.612869	0.924449	0.926129	0.925037

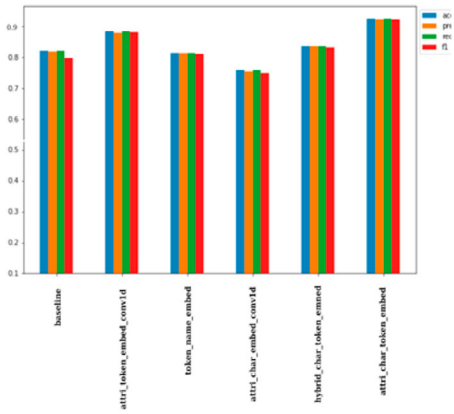

Fig. 10. Experimental results

5. Conclusions and Future Work

The research conducted a comprehensive examination of the unique attributes associated with individual names, delving into potential sources of variation and error inherent in name data. In addition to this, this research provided a thorough overview of both phonetic encoding and pattern-matching techniques employed in name matching. Experimental results obtained from a diverse range of real-world datasets highlighted the absence of a one-size-fits-all solution. When selecting a name-matching technique, researchers must carefully consider the specific characteristics of the dataset being analyzed, as well as the computational demands associated with each technique.

Individual name matching poses significant challenges, necessitating further investigation into the attributes of name data and matching methodologies. A deeper understanding is required to discern why certain techniques outperform others and to determine which techniques are most suitable for various data types. To gain greater insights, it is imperative to conduct a more granular analysis of error types and their distribution, shedding light on how particular error categories impact the performance of matching techniques.

The utilization of the Name2Vec method, in conjunction with character-based properties extracted from name strings, as similarity indices for comparing two strings has exhibited promising results. Through rigorous testing, this approach has demonstrated enhanced performance, rendering name matching more efficient and mitigating misclassification issues associated with existing methods tailored for Vietnamese names.

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